

PROJECT TASK

UK Rail Delay and Passenger Experience
Analytics

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PROJECT GOAL

Build a data analytics project that studies rail delays and passenger experience in the UK using real-world rail performance data. The project should explore patterns in delays, cancellations, route performance, operator performance, and passenger impact, then present clear insights through analysis, visualisations, and a dashboard.

PROJECT CONTEXT

Rail transport plays an important role in the UK, and delays affect thousands of passengers every day. These delays can impact work, education, travel plans, and overall customer satisfaction. Train operators and transport authorities need to understand where delays happen most often, which routes or operators perform poorly, and how trends change across time.

This project focuses on analysing UK rail performance data to uncover useful patterns and communicate meaningful insights. It should feel like a real analytics case study rather than only a notebook with charts. The aim is to combine data cleaning, exploration, business understanding, and storytelling into one portfolio project.

The project should be built in a way that demonstrates strong data analysis skills, clear thinking, and practical communication. It should also leave room for future extension into machine learning, such as delay prediction or demand forecasting, but the main focus for now is analytics.

YOUR MISSION

Your job is to investigate rail delay data and answer questions such as:

- which operators perform best and worst
- which stations or routes face the most delay issues
- how delay patterns change by time period
- where cancellations are becoming a bigger problem
- what trends may affect passenger experience

Your goal is not just to make graphs. Your goal is to turn raw rail data into a story that helps someone make decisions.

By the end of the project, anyone should be able to look at your GitHub and quickly understand:

- what problem you explored
- how you worked with real data
- what insights you found
- how clearly you communicated them

SUGGESTED MILESTONES / PHASES

Phase 1. Understand the Problem and Find the Data

- understand the rail analytics problem
- find and inspect real UK rail data
- decide the focus of the project
- define key questions to answer

Phase 2. Prepare the Data

- collect datasets
- clean and organise the data
- handle missing values and inconsistent formats
- prepare data for analysis

Phase 3. Explore and Analyse

- perform exploratory data analysis
- identify trends, patterns, and unusual behaviour
- compare operators, routes, stations, or time periods
- create useful metrics and summaries

Phase 4. Build the Story and Dashboard

- design charts and visual outputs
- build a dashboard in Tableau, Power BI, or a similar tool
- turn findings into a clear story
- highlight insights that matter for passenger experience

Phase 5. Finalise the Portfolio Project

- prepare GitHub structure
- write README and project explanation
- include screenshots and insights
- suggest future ML extensions

TASKS

TASK 1: Understand the Rail Analytics Problem

Before working with the data, the first step is to understand the real-world problem this project is trying to solve.

This is not simply about delays as numbers. Behind every delayed train there is a passenger experience issue, an operational issue, and a business impact.

Imagine this project is being prepared for a transport analytics team that wants to improve rail performance and customer satisfaction across the UK.

Your job in this task is to think like a data analyst and understand what questions matter.

WHAT TO DO

Write a short project understanding note that explains:

- what the project is about
- why rail delays matter
- who would use this analysis
- what business or operational problem this project is solving
- how passenger experience is affected by delays and cancellations

Think about the possible stakeholders who may use this analysis:

- rail operators
- transport managers
- passenger service teams
- station operations teams
- customer experience teams
- regulators or reporting teams

QUESTIONS TO THINK ABOUT

Start by thinking about questions such as:

- Are delays getting worse over time?
- Which operators perform poorly?
- Are some routes consistently delayed?
- Which time periods are most affected?
- Do cancellations impact certain regions more?
- Which issues matter most for passengers?

TASK 2: Find and collect the Data

Now that the problem is clear, the next step is to work with **real UK rail data**. This is one of the strongest parts of the project because it uses real-world datasets that are highly relevant.

The main recommended source is the **Office of Rail and Road (ORR) Data Portal**, which provides official rail performance data for Great Britain.

This includes:

- punctuality data
- delay performance
- cancellations
- operator performance
- station and route statistics

This makes the project highly credible.

WHAT TO DO

Collect one or more datasets that support your project goals. Suggested data sources:

- ORR rail performance data
- station-level performance data
- cancellation statistics
- route or operator reports

You may combine multiple files if useful.

THINGS TO CHECK WHEN COLLECTING DATA

Before moving forward, inspect the raw data carefully. Look at:

- file type
- number of rows and columns
- date fields
- operator names
- route / station identifiers
- missing values
- unclear field names

QUESTIONS TO THINK ABOUT

- What does each column represent?
- Is the data monthly, quarterly, or yearly?
- Does it cover all operators?
- Are there multiple datasets that need merging?
- Are time formats consistent?

OUTPUT EXPECTED

A raw data folder in your project, for example:

```
data/  
  raw/
```

Along with a short note explaining:

- where the data came from
- why you selected it
- what time range it covers
- what key fields are available

TASK 3: Define the Questions and Metrics

In this task, you should decide what exactly you want your analysis to answer. These questions will guide your EDA, dashboard, and final recommendations. Think of this as deciding what story the data should tell.

WHAT TO DO

Define the main analytical questions for the project. Start with 5 to 8 strong business questions. Here are some example directions:

Performance questions

- Which train operators have the highest delay rates?
- Which operators perform best?
- How have delays changed over time?

Route / station questions

- Which stations experience the most frequent delays?
- Are certain routes more affected than others?
- Are some regions consistently underperforming?

Time-based questions

- Which months or quarters have the worst performance?
- Are weekday delays worse than weekend delays?
- Are there seasonal patterns?

Passenger experience questions

- Which routes have the highest cancellation impact?
- Where are passengers most likely to face disruption?
- Which operators show improving or declining trends?

DEFINE KEY METRICS

Now define the KPIs or metrics you want to track. Suggested metrics include:

- total delayed services
- total cancelled services
- delay percentage
- cancellation percentage
- operator performance score
- average delay trend
- route-level disruption frequency
- punctuality rate

OUTPUT EXPECTED

Create a short section in your notes that includes:

Analytical questions

A list of key questions you want the data to answer

Metrics

A list of KPIs you will use in your analysis and dashboard

TASK 4: Plan the Project Structure

Before jumping into coding or building dashboards, it is important to organise the project properly. A well-structured project makes your work easier to understand, easier to maintain, and much stronger for GitHub. Think of this as designing the workspace for your analysis. Anyone looking at the project should be able to open the repository and immediately understand where everything is.

WHAT TO DO

Create a clear folder structure for the project. A recommended structure is:

```
uk-rail-delay-passenger-analytics/  
├── data/  
│   ├── raw/  
│   └── cleaned/  
├── notebooks/  
├── dashboards/  
├── visuals/  
├── docs/  
└── README.md
```

SUGGESTED MEANING OF EACH FOLDER

data/raw

This should contain the original downloaded files exactly as received.

Do not manually edit raw files.

This helps maintain traceability.

data/cleaned

This should contain processed datasets after cleaning. Examples:

- renamed columns
- fixed dates
- filtered records
- merged files
- missing values handled

notebooks

This should contain analysis notebooks. Suggested notebook flow:

- 01_data_understanding.ipynb
- 02_data_cleaning.ipynb
- 03_eda.ipynb
- 04_operator_analysis.ipynb
- 05_dashboard_prep.ipynb

This numbering makes the flow very clear.

dashboards

This should contain Tableau / Power BI files. For example:

- .twb
- .pbix

visuals

Store exported charts and screenshots here. This is useful for:

- GitHub README
- dashboard previews
- project presentation

docs

This can include:

- project notes
- business questions
- KPI definitions
- summary insights

WHAT TO THINK ABOUT

- Where should cleaned datasets live?
- How should notebook naming show workflow order?
- How can someone reviewing your GitHub understand the project journey?

OUTPUT EXPECTED

Create the folder structure in your local project.

This should be committed to GitHub from the beginning.

TASK 5: Clean and Prepare the Data

Real-world data is rarely ready for analysis immediately. This task is where you demonstrate practical data analytics skills. For any analytics job, this is often one of the most valued parts because it shows you can work with messy, real datasets.

WHAT TO DO

Take the raw rail dataset and prepare it for analysis.

The cleaned data should be reliable, consistent, and easy to use for further analysis.

KEY CLEANING AREAS

1. Understand the dataset first

Before changing anything, inspect the dataset.

Check:

- total rows and columns
- data types
- sample values
- null values
- duplicate rows
- unusual categories

Useful things to inspect:

- operator names
- date fields
- route / station fields
- delay counts
- cancellation counts

2. Handle missing values

Check which columns contain missing values.

Think carefully:

- can these rows be removed?
- should values be filled?
- should they be marked as unknown?

For example:

- missing station names
- missing route values
- missing operator code

- missing date

Do not remove rows blindly.

3. Standardise column names

Make the dataset easier to work with. For example:

Instead of:

Train Operator Name

use:

train_operator_name

This improves readability.

4. Convert dates properly

This is extremely important for time-based analytics. Make sure date columns are converted to proper datetime format.

For example:

- month
- quarter
- year
- date of service

Once converted, you can create time trends easily.

5. Check duplicates

Look for duplicate records. Questions to ask:

- are these real duplicates?
- are they repeated reporting periods?
- do multiple rows represent the same operator and date?

Only remove duplicates after understanding them.

6. Create helpful derived columns

This is where the analysis becomes stronger. Examples:

- year
- month
- quarter
- delay_rate
- cancellation_rate

For example:

```
delay_rate = delayed_services / total_services
```

These columns will be extremely useful later.

7. Save cleaned data

Once cleaned, save the processed dataset.

Suggested location:

```
data/cleaned/
```

Example filename:

```
rail_performance_cleaned.csv
```

QUESTIONS TO THINK ABOUT

- Which fields are critical for the analysis?
- What columns should be derived now instead of later?
- How will the cleaned data support dashboard creation?

OUTPUT EXPECTED

By the end of this task, you should have:

- cleaned dataset
- documented cleaning steps
- saved processed file
- notebook showing cleaning workflow

This notebook should be easy for someone else to follow.

TASK 6: Explore the Data

Now that the dataset is cleaned and ready, the next step is to understand what the data is trying to tell you. This task is all about exploration. Before jumping into conclusions, spend time looking at patterns, distributions, trends, and unusual values. Think of this as “getting to know the data”. The purpose here is not only to create graphs, but to start identifying possible stories and business insights.

WHAT TO DO

Perform an Exploratory Data Analysis (EDA) of the cleaned rail dataset. Start by understanding the overall shape and structure of the data.

AREAS TO EXPLORE

1. Basic overview

Start with high-level questions such as:

- how many rows and columns are there
- what time period does the data cover
- how many train operators are included
- how many routes or stations are present

This gives a broad understanding of scope.

2. Distribution analysis

Study the spread of important metrics.

Examples:

- delayed services
- total services
- cancellation counts
- delay rate
- cancellation rate

Useful visuals:

- histograms
- box plots
- distribution plots

This helps answer:

- are delays generally low but with extreme spikes?
- are cancellations rare or common?

3. Time-based exploration

This is one of the most important parts. Look at trends over time.

Examples:

- delays by month
- delays by quarter
- cancellations by year
- seasonality

Try visuals such as:

- line charts
- moving averages
- month-wise trend plots

This will help reveal:

- improving performance
- worsening periods
- recurring seasonal issues

4. Operator exploration

Compare operators. Examples:

- total delayed services by operator
- average delay rate by operator
- cancellation rate by operator

5. Outlier detection

Look for unusual spikes.

Examples:

- sudden large delay jumps
- extreme cancellation months
- operators with unusual performance

Think about:

- data issues
- real operational events
- strikes or weather disruptions

OUTPUT EXPECTED

Create an EDA notebook, for example:

This notebook should include:

- observations
- visualisations
- written notes below charts
- early patterns noticed

IMPORTANT NOTE

Do not just create charts. Below every chart, write:

- what it shows
- what stands out
- what question it raises

For example:

Delay rates sharply increased in Q4 across multiple operators, which may indicate seasonal congestion or service disruptions.

TASK 7: Analyse Delays, Cancellations, and Performance

Now the focus is **business insight and comparison**. This is where the project begins to answer real questions that matter for transport performance and passenger experience.

Think of this as the main analytical core of the project.

WHAT TO DO

Analyse key performance areas and compare them in a structured way. Focus on three main pillars:

- delays
- cancellations
- operator performance

PART 1. DELAY ANALYSIS

Start by deeply analysing delays.

Questions to answer

- which operators have the highest delay rates?
- which periods are worst?
- are delays improving over time?
- which routes are most affected?

Useful charts:

- operator comparison bar charts
- monthly line trends
- heatmaps by month and operator
- top delayed routes

Try creating KPIs such as:

```
average_delay_rate  
worst_operator_delay_rate  
best_operator_delay_rate
```

PART 2. CANCELLATION ANALYSIS

Now focus on service cancellations. This is very important for passenger experience.

Questions:

- which operators cancel the most services?
- are cancellations rising?
- do cancellations follow the same delay pattern?
- which time periods are worst?

Useful metrics:

- total cancellations

- cancellation %
- monthly cancellation trend

Suggested visual:

- stacked bar charts
- trend lines
- operator comparison charts

PART 3. OPERATOR PERFORMANCE RANKING

This is a very strong portfolio section. Create a comparative view of operator performance.

Possible ranking metrics:

- delay %
- cancellation %
- punctuality %
- combined service score

For example, create something like:

```
performance_score =
0.6 * punctuality_rate
- 0.4 * cancellation_rate
```

QUESTIONS TO THINK ABOUT

- which operators consistently underperform?
- which ones are improving?
- are there major gaps between operators?
- what would this mean for passengers?

OUTPUT EXPECTED

A separate notebook such as:

```
04_operator_performance_analysis.ipynb
```

This notebook should clearly answer the major business questions.

Each section should contain:

- analysis
- visualisation
- written insight
- short recommendation

Example:

Operator X shows the highest cancellation rate across the last four quarters and may require operational investigation.

TASK 8: Study Passenger Experience Patterns

Up to this point, the focus has been mainly on delays, cancellations, and operator performance. Now we take the next step and think about how these issues affect passengers. This is what makes the project feel more human and more relevant to real-world transport analytics.

- Remember, a delayed service is not just an operational issue. It affects:
- work commuters
- students
- families travelling
- airport connections
- customer satisfaction

This task is about understanding the impact behind the numbers.

WHAT TO DO

Use the existing rail performance data to explore patterns that may reflect passenger experience and service disruption. Think beyond raw delay counts.

Try to answer questions such as:

- when are passengers most likely to face disruption
- which operators may cause the highest passenger inconvenience
- are peak travel periods more affected
- are there repeated seasonal issues

AREAS TO EXPLORE

1. Peak disruption periods

Study whether certain time periods consistently show worse performance.

Examples:

- specific months
- winter season
- holiday periods
- weekday vs weekend
- morning and evening peak travel windows, if available

This helps understand when passengers are most impacted.

2. Operator experience comparison

Compare operators from a passenger point of view.

Instead of only:

| *highest delay %*

think in terms of:

where would a passenger most likely experience disruption?

Possible metrics:

- disruption frequency
- delay trend stability
- cancellation likelihood

3. Passenger impact scoring

This is a very strong portfolio idea. Try creating a simple custom metric such as:

```
passenger_impact_score =  
0.7 * delay_rate  
+ 0.3 * cancellation_rate
```

You can adjust the logic. The purpose is to create a **passenger-centric KPI**.

This shows analytical creativity.

4. Trend storytelling

Try writing insights such as:

Passenger disruption appears to increase significantly during winter months, likely due to seasonal weather conditions and increased operational pressure.

This makes the project stronger than simple chart plotting.

OUTPUT EXPECTED

A notebook section or separate notebook such as:

```
05_passenger_experience_analysis.ipynb
```

This should include:

- insights
- visual comparisons
- business-oriented observations
- passenger-focused metrics

TASK 9: Create Visualisations

The goal is not to create as many charts as possible. The goal is to create **clear, decision-friendly visuals**. A good analyst tells a story through charts. Each visual should help answer a question.

WHAT TO DO

Create strong visualisations that communicate the most important insights from your analysis. Think carefully about chart choice.

SUGGESTED VISUALISATIONS

1. Delay trends over time

Use:

- line chart
- moving average trend

Best for:

- monthly delays
- quarterly delay rates
- operator trends

2. Operator comparison

Use:

- bar charts
- sorted comparison charts

Best for:

- top / bottom operators
- cancellation rate comparison
- performance ranking

3. Heatmaps

Very strong for time-based patterns.

Examples:

- operator vs month heatmap
- delays by quarter
- seasonal impact

These look very professional in portfolios.

4. KPI cards

Examples:

- total delays
- total cancellations
- best operator
- worst operator
- average delay %

These will be useful for the dashboard.

5. Insight visuals

Choose visuals that help communicate findings clearly.

Examples:

- top disrupted routes
- disruption trend by season
- passenger impact ranking

IMPORTANT RULE

Every chart should answer one question. Below each chart, write:

- what the chart shows
- key observation
- why it matters

Example:

Operator A shows a consistent upward trend in cancellations over the last three quarters, suggesting declining service stability.

TASK 10: Build the Dashboard

This is the final presentation layer of the project. This is where all your work comes together. The dashboard should feel like something that a transport manager or analyst team could actually use.

WHAT TO DO

Build an interactive dashboard that presents the key insights from the project. The dashboard should allow someone to quickly understand:

- rail performance
- disruption trends
- operator comparisons
- passenger impact

SUGGESTED SECTIONS

Section 1. KPI summary

Top cards showing:

- total delayed services
- total cancellations
- average delay rate
- best operator
- worst operator

Section 2. Trend view

Add line charts showing:

- delay trend over time
- cancellation trend over time

Section 3. Operator comparison

- ranking chart
- delay %
- cancellation %
- performance score

Section 4. Passenger experience

- passenger impact score
- disruption trend
- seasonal risk

Section 5. Filters

Add filters such as:

- operator
- year / quarter
- route if available

DESIGN GUIDANCE

Keep it clean and professional. Avoid clutter.

Use:

- clear spacing
- readable labels
- consistent font sizes
- intuitive filter placement

Think about how a recruiter or hiring manager would scan it quickly.

OUTPUT EXPECTED

A dashboard file in:

dashboards/

Examples:

- Tableau workbook
- Power BI file
- screenshots exported into visuals folder

TASK 11: Write Insights and Recommendations

A data analyst's role is not just to create charts. The real value comes from explaining what the data means and what should be done next. This task is about turning your analysis into clear business insights and actionable recommendations.

Think of this as writing the "so what?" section of the project.

WHAT TO DO

Write a structured summary of your findings. This should not be a technical explanation of code. Instead, it should focus on:

- what patterns you found
- what trends stand out
- what risks or opportunities exist
- what recommendations you would give to a transport team

SUGGESTED STRUCTURE

1. Key Findings

Summarise the most important insights from the analysis.

Examples:

- the worst-performing operators
- periods with highest delay rates
- seasonal disruption patterns
- routes with repeated service issues
- passenger disruption hotspots

Example style:

Delay rates increased significantly during Q4 across multiple operators, with winter months showing the highest disruption levels.

2. Business Impact

Now explain why these findings matter.

Examples:

- commuter inconvenience
- reduced passenger satisfaction
- operational inefficiency
- financial or service reputation risk

Example:

High cancellation rates during weekday morning periods may significantly impact daily commuters and workplace productivity.

3. Recommendations

This is extremely important. Give realistic recommendations.

Examples:

- focus operational reviews on underperforming operators
- investigate repeated seasonal bottlenecks
- improve staffing during peak disruption periods
- strengthen service contingency plans

Example:

Operators with repeated winter disruption should be reviewed for route resilience and weather preparedness.

OUTPUT EXPECTED

A written insights summary inside:

```
docs/  
project_summary.md
```

This should also be included in the README later.

TASK 12: Organise the GitHub Project Properly

Now it is time to make the repository professional and easy to review. This is extremely important because GitHub is often the first thing recruiters see. The goal is to make the project easy to navigate. Someone should understand the full workflow within a few minutes.

WHAT TO DO

Review the full folder structure and organise everything clearly. Suggested final structure:

```
uk-rail-delay-passenger-analytics/  
├── data/  
│   ├── raw/  
│   └── cleaned/  
├── notebooks/  
├── dashboards/  
├── visuals/  
├── docs/  
└── README.md
```

IMPORTANT CHECKS

1. Notebook naming

Make sure notebook names follow workflow order.

Example:

```
01_problem_understanding.ipynb  
02_data_collection.ipynb  
03_data_cleaning.ipynb  
04_eda.ipynb  
05_operator_analysis.ipynb  
06_passenger_experience.ipynb
```

This makes the story easy to follow.

2. Clear visuals folder

Store all exported visuals and dashboard screenshots in one place.

This is important for the README.

3. Clean notebook presentation

Make sure notebooks include:

- markdown headings
- chart explanations
- short insight notes

Avoid long code-only notebooks.

4. Remove unnecessary clutter

Remove:

- temporary files
- repeated CSV versions
- unnecessary notebook outputs
- unused visuals

The repository should feel polished.

TASK 13: Write the README and Documentation

A strong README can make the difference between someone exploring the project and someone leaving immediately.

Think of this as the project landing page.

WHAT TO DO

Create a professional README file. The README should clearly explain the project from start to finish.

SUGGESTED README STRUCTURE

1. Project Title

2. Project Overview - A short description of what the project is about.

3. Business Problem - Explain the problem clearly.

4. Dataset

Mention:

- source
- time period
- important columns
- why the dataset is relevant

5. Tools Used

6. Project Workflow

Summarise steps:

- data collection
- cleaning
- EDA
- performance analysis
- dashboard
- insights

7. Key Insights - Include the main findings.

8. Dashboard Preview (*Very important*) - Add screenshots from the dashboard.

9. Recommendations - Include business recommendations.

TASK 14: Suggest Future Extensions and ML Ideas

BONUS TASK 1: DELAY PREDICTION MODEL

This is the most natural ML extension for the project. The goal is to predict the likelihood of service delays based on historical rail performance data.

This would be highly relevant for transport analytics and operational planning.

Possible problem statement

Use historical data to predict whether a train operator or route is likely to experience high delays in the next reporting period.

Possible target

Examples:

- delay rate for next month
- delay category such as low / medium / high
- probability of severe disruption

Suggested features

- operator
- month / quarter
- historical delay trend
- cancellation trend
- seasonal indicators
- route / station if available

Suggested models

- linear regression
- decision tree
- random forest
- XGBoost, later as an advanced option

Why this is valuable

This demonstrates:

- forecasting mindset
- supervised learning
- feature engineering

Very strong for data science roles.

BONUS TASK 2: CANCELLATION RISK PREDICTION

This is another excellent business use case. The goal is to predict the likelihood of cancellations for future periods. This could help operators plan resources and improve reliability.

Possible objective

Predict whether a service, route, or operator is at high risk of cancellation in an upcoming period.

Example framing

Binary classification:

0 = normal risk 1 = high cancellation risk
--

Suggested features

- previous delay rates
- operator performance score
- seasonal period
- route risk history
- disruption trend

Suggested models

- logistic regression
- random forest classifier
- gradient boosting

BONUS TASK 3: ANOMALY DETECTION FOR UNUSUAL DISRUPTIONS

The idea is to identify unusual spikes in delays or cancellations. This can simulate real-world alerting systems.

Example questions

- which months show abnormal disruption
- which operators had unusual performance drops
- are there unexpected route-level spikes

Suggested techniques

- z-score based anomaly detection
- isolation forest
- time-series anomaly detection

This is highly relevant for operations analytics.

BONUS TASK 4: PASSENGER DISRUPTION FORECASTING

This is one of the strongest future extensions. Use time-series methods to forecast disruption trends. This is extremely relevant for transport and demand planning roles.

Example goal

Forecast:

- delays for next quarter
- cancellation trend
- passenger impact score

Suggested approaches

- moving average
- ARIMA
- Prophet
- XGBoost with lag features later

This makes the project highly professional.

BONUS TASK 5: GEOSPATIAL / REGIONAL ANALYSIS

This is a fantastic visual and analytical upgrade. If route or station location data is available, the project can include map-based analysis.

Possible extensions

- region heatmaps
- station disruption hotspots
- route performance clusters

Tools

- Python mapping libraries
- Tableau maps
- Power BI map visuals

Very strong for storytelling.

BONUS TASK 6: INTERACTIVE WEB DASHBOARD EXTENSION

This is an excellent portfolio booster. Later the dashboard can be turned into a lightweight analytics application.

Possible tools:

- Streamlit
- Plotly Dash

Possible features

- interactive operator selection
- time-based filtering
- prediction panels
- live KPI cards
- future risk scores

BONUS TASK 7. ADVANCED BUSINESS QUESTIONS

Future questions may include:

- which operators are improving over time
- which routes need operational intervention
- where should resources be allocated
- which seasons create the highest passenger risk